



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-12	Team No-12	Project Title-Job Portal with Resume Parsing & Matching
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1. Abstract

Header Details

Project Title: Job Portal with Resume Parsing & Matching (Intelligent ATS with Polyglot Microservices)

1. Abstract

Traditional recruitment workflows are largely manual, leading to significant delays in processing high application volumes and matching candidates accurately with job requirements. This project presents an Intelligent Job Portal and Applicant Tracking System (ATS) built on a resilient, polyglot microservices architecture to automate resume parsing, skill matching, and the hiring lifecycle.

The system employs a polyglot persistence model: a relational database (PostgreSQL/MySQL) handles structured core portal entities and recruiter analytics; MongoDB manages candidate activity feeds and logs; and a Vector Database (pgvector/Elasticsearch) performs semantic skill similarity scoring using NLP embeddings.

Backend operations are decoupled behind a centralized FastAPI API Gateway handling JWT authentication and routing. A Node.js/Express service manages real-time application pipelines, recruiter workflows, and interview notifications via WebSockets and Redis caching. A Python/FastAPI microservice runs the Natural Language Processing (NLP) pipeline to parse uploaded resumes and score candidate suitability. The polyglot ecosystem is containerized with Docker, prepared for Kubernetes deployment, monitored with Prometheus and Grafana, and delivered via automated CI/CD pipelines.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090010	komali varnika	
2520090006	K. Aasritha sai Lakshmi,	
2520090031	N.Divya sree	

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

- Frontend: React.js, Tailwind CSS / Material-UI, Axios
- Backend Frameworks & Runtimes: Node.js, Express.js, Python (FastAPI), Pydantic
- Databases & Search Engines:
 - Relational: PostgreSQL / MySQL (with SQLAlchemy / Sequelize)
 - NoSQL / Document: MongoDB (Mongoose / Motor)
 - Vector Search: pgvector / Elasticsearch / Pinecone
 - In-Memory & Caching: Redis
- NLP & Parsing Tools: Python spaCy, NLTK, pdfplumber / PyPDF2, Sentence-Transformers
- Real-Time Communication & Auth: Socket.io, JSON Web Tokens (JWT), OAuth2
- DevOps, Cloud & Containerization: Docker, Docker Compose, Kubernetes (Ingress, Pods, Services)
- Observability & CI/CD: Prometheus, Grafana, OpenTelemetry, GitHub Actions (CI/CD & security scanning)

Guide Name and Signature (remarks if any) _____